

	Vdd (V)	1		$I_{dn} = I_{dp}$	
	Vtn (V)	0.3		$K_n/K_p = 2$	
	$ V_{tp} $ (V)	0.3		$W_p/W_n = r$	
	$I_{dn} = k_n W_n (V_m - V_{tn})$			$I_{dp} = K_p W_p (V_{dd} - V_m - V_{tp})$	
	$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - V_{tp})$				
	$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - V_{tp})$				
	$2 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - V_{tp})$				
	$2 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$				
	$2 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$				
	$2 V_m - 0.6 = r (0.7 - V_m)$				
	$2 V_m - 0.6 = 0.7r - r V_m$				
	$2 V_m + r V_m = 0.7r + 0.6$				
	$(2+r) V_m = 0.7r + 0.6$				
	$V_m = (0.7r + 0.6) / (2+r)$				
	Design	Wn	Wp	r	Vm
	A	1	1	1	0.4333333333
	B	1	2	2	0.5
	C	1	3	3	0.54
	Vdd (V)	1		$I_{dn} = I_{dp}$	
	Vtn (V)	0.3		$K_n/K_p = 1.6$	
	$ V_{tp} $ (V)	0.3		$W_p/W_n = r$	
	$I_{dn} = k_n W_n (V_m - V_{tn})$			$I_{dp} = K_p W_p (V_{dd} - V_m - V_{tp})$	
	$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - V_{tp})$				
	$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - V_{tp})$				
	$1.6 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - V_{tp})$				
	$1.6 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$				
	$1.6 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$				
	$1.6 V_m - 0.6 = r (0.7 - V_m)$				
	$1.6 V_m - 0.6 = 0.7r - r V_m$				
	$1.6 V_m + r V_m = 0.7r + 0.6$				
	$(1.6+r) V_m = 0.7r + 0.6$				
	$V_m = (0.7r + 0.6) / (1.6+r)$				

	Design	Wn	Wp	r	Vm
	A	1	1	1	0.5
	B	1	2	2	0.5555555556
	C	1	3	3	0.5869565217
	Vdd (V)	1		Idn = Idp	
	Vtn (V)	0.3		Kn/Kp = 2.6	
	Vtp (V)	0.3		Wp/Wn = r	
	Idn = kn Wn (Vm-Vtn)		Idp = KpWp(Vdd-Vm- Vtp)		
	$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - V_{tp})$				
	$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - V_{tp})$				
	$2.6 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - V_{tp})$				
	$2.6 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$				
	$2.6 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$				
	$2.6 V_m - 0.6 = r (0.7 - V_m)$				
	$2.6 V_m - 0.6 = 0.7r - r V_m$				
	$2.6 V_m + r V_m = 0.7r + 0.6$				
	$(2.6 + r) V_m = 0.7r + 0.6$				
	$V_m = (0.7r + 0.6) / (2.6 + r)$				
	Design	Wn	Wp	r	Vm
	A	1	1	1	0.3611111111
	B	1	2	2	0.4347826087
	C	1	3	3	0.4821428571
	Design	Nominal	Slow Nmos Fast Pmos	Fast Nmos Slow Pmos	Vm Spread
	A	0.4333333333	0.5	0.3611111111	0.1388888889
	B	0.5	0.5555555556	0.4347826087	0.1207729469
	C	0.54	0.5869565217	0.4821428571	0.1048136646
	Design C has the smallest Vm spread				
	Vdd/2:	0.5			
	Design B is closest to Vdd/2 under nominal conditions.				